A pre-registered test of cross-fitted stacking with a novelty-damped local residual correction on the Matbench composition tasks

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Abstract

We test whether a local residual correction confirmed in an earlier chemistry study transfers to composition-based materials property prediction. The pipeline has two stages:

- cross-fitted stacking of diverse experts;
- a novelty-damped local out-of-fold residual correction.

The test uses the four composition tasks of Matbench v0.1 (steels, expt_gap, expt_is_metal, glass) under a strict protocol:

- only the official folds are used;
- every model, hyperparameter and design choice is made by cross-validation inside each training fold;
- the analysis plan was written, and the code frozen by SHA-256 hash, before any test fold was scored;
- every task, fold and comparison is reported.

The simplex-constrained stack never lost to an equal-weight ensemble and improved calibration. Its gains over the best single expert were small (largest on glass). The local correction improved one of the four tasks under the pre-registered rule. On expt_gap the mean absolute error fell from 0.3264

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to 0.3240 eV (-0.71%, 5/5 folds, Holm-adjusted $p = 0.005$). It had no detectable effect on steels (-0.39%), and its own nested selection declined it on the two classification tasks. Stratifying by distance to the nearest training composition shows that the correction hurts for near-duplicate compositions, where the deployed stack has already absorbed the neighbours' errors. A fold-matched variant designed to remove this double counting did not help. Among the leaderboard entries the method would rank 3, 2, 2 and 2 on steels, expt_gap, expt_is_metal and glass. It does not beat the best entry on any task. We also show that most leaderboard classification scores are label-based balanced accuracies, which are not comparable with probability ROC-AUCs.

Keywords: materials informatics, composition-based property prediction, Matbench, stacking, residual correction, nested cross-validation, pre-registration

1. Introduction

Composition-only property models remain the workhorse of materials screening whenever the crystal structure is unknown, and the Matbench v0.1 suite (Dunn et al., 2020) has become the common yardstick for them. Four of its tasks are composition-only and comparatively small: the yield strength of steels (312 alloys), the experimental band gap (4,604 compounds), the experimental metal/non-metal label (4,921) and glass-forming ability (5,680). On these tasks the leaderboard (Matbench developers, 2026) is led by quite different model families: automated machine-learning pipelines on steels, feature-selected neural networks (MODNet, De Breuck et al., 2021b) on glass, and the Darwin entry on the two experimental band-gap tasks. The composition models we are aware of either fit one global predictor or combine predictors with weights that are the same for every composition (as automated-ML ensembles and stacked models do) (De Breuck et al., 2021b; Wang et al., 2021; Goodall and Lee, 2020; Gouvêa et al., 2025; Zou et al., 2025).

Errors of such models are rarely uniform over composition space. Chemical families share systematic biases, and a residual that is large for one alloy is often large for its neighbours. In geostatistics this observation underlies regression kriging, where a trend model is followed by interpolation of its residuals (Hengl et al., 2007); in quantum chemistry, Δ-learning corrects a cheap baseline with a learned residual (Ramakrishnan et al., 2015). In earlier

work on chemical property prediction (SIG-Net, Gurram, 2026) we combined a cross-fitted stack of diverse experts (Wolpert, 1992; Breiman, 1996; van der Laan et al., 2007) with a *novelty-damped local out-of-fold (OOF) residual correction*: each prediction is shifted by a kernel-weighted average of the OOF residuals of its nearest training neighbours, and the shift is damped where the query is far from all of them. That correction was confirmed in a pre-registered replication.

Here we ask whether the same construction transfers to composition-based materials property prediction, under the strictest protocol we could devise on a public benchmark: (i) only the official Matbench folds are used; (ii) every choice (expert hyperparameters, stacking weights, correction strength, neighbourhood size, bandwidth, damping and kernel space) is made by cross-validation inside each training fold; (iii) the analysis plan, endpoints and tests were written and the code was frozen by cryptographic hash before any official test fold was predicted; and (iv) every task, fold and arm is reported. We also test a variant suggested by a structured ideation-and-review process (Section 3.5): a *fold-matched* correction that removes a double-counting term which arises because the deployed stack has already been trained on the neighbours whose residuals it borrows.

2. Related work

Composition models on Matbench. Magpie descriptors with random forests (Ward et al., 2016) and the Automatminer reference (Dunn et al., 2020) established the baselines; MODNet (De Breuck et al., 2021b,a) added relevance-redundancy feature selection and joint learning for small data; CrabNet (Wang et al., 2021) and Roost (Goodall and Lee, 2020) learn representations directly from stoichiometry; MatterVial fuses latent, symbolic and descriptor features into a MODNet model (Gouvêa et al., 2025); and tabular foundation models such as TabPFN (Hollmann et al., 2025) are now competitive on small tabular sets. None of these adapts its prediction to the residual structure of nearby training compositions.

Ensembles and stacking for materials. Stacked ensembles have been used for experimental band gaps with DFT-trained first-layer models (Kauwe et al., 2020), for thermodynamic stability from electron-configuration and other composition models (Zou et al., 2025), and for alloy mechanical properties on single datasets with random splits (e.g. Zhao et al., 2025). A mixture

of experts over pretrained extractors addresses data scarcity (Chang et al., 2022), and an "ensemble of experts" built from networks pre-trained on related polymer properties was recently proposed in this journal (Miccio, 2025); the latter is transfer learning from SMILES and contains no stacking or residual stage.

Residual correction. Regression kriging (Hengl et al., 2007) and Δ-learning (Ramakrishnan et al., 2015) are the closest classical analogues of a local residual correction. A recent preprint corrects a convex neural stack by global kernel-ridge regression of its residuals in operator learning (Shmalo, 2026). None of these uses OOF residuals with novelty damping in composition space, or evaluates on Matbench folds; the method itself, however, is not new in its components, and our contribution is the pre-registered transfer test, the fold-matched variant and the analysis of where the correction helps or hurts. A structured prior-art check (details in the Supplementary Material) rated the overlap as medium: every component has ancestors, but we found no work that combines them on these tasks.

3. Methods

3.1. Data, splits and features

We use the four composition tasks of Matbench v0.1 through the `matbench` package (`MatbenchBenchmark`), with the official five folds; nothing is re-split. Each composition is described by label-free features computed with `matminer` (Ward et al., 2018) and `pymatgen` (Ong et al., 2013): 132 Magpie element-property statistics (Ward et al., 2016), 103 element fractions, and 21 further descriptors (stoichiometric norms, valence-orbital fractions (Meredig et al., 2014), ionic character and charge-balance flags, band centre and atomic-orbital HOMO/LUMO estimates). A handful of compositions (0, 14, 14 and 21 rows in the four tasks) make one of the extra featurisers fail; these values are imputed with the feature's dataset-wide median, which uses no labels.

3.2. Experts and cross-fitting

Inside an official training fold T_k the training rows are split into $J = 5$ inner folds $P_1, \ldots, P_J$ (seed 0; stratified for classification). Every expert e is fitted J times on $T_k \setminus P_m$ to produce OOF predictions $\hat{f}_e^{(-m(i))}(x_i)$, and once on all of T_k to predict the test rows. The expert pool is deliberately

heterogeneous and CPU-feasible: LightGBM (Ke et al., 2017) on all features (for expt_gap also an L1-objective copy), random forests (Breiman, 2001), extremely randomised trees (Geurts et al., 2006), kernel ridge regression on element fractions and on standardised Magpie features (RBF kernel, regularisation and width chosen by an internal three-fold grid search), RBF support-vector classifiers on the same two spaces for the classification tasks, and a two-layer perceptron (Pedregosa et al., 2011). Regression predictions are clipped to the training-target range.

3.3. Simplex stack

The global stack is $s(x) = \sum_e w_e \hat{f}_e(x)$ with $w_e \geq 0$ and $\sum_e w_e = 1$, fitted on the OOF predictions. For regression the weights minimise the mean absolute error (a linear programme, matching the Matbench metric); for classification they minimise the Brier score. To obtain OOF *stack* predictions $s^{\text{oof}}(x_i)$ for the training rows, the weights used for inner fold m are fitted without the rows of P_m.

3.4. Novelty-damped local OOF residual correction

With OOF residuals $r_j = y_j - s^{\text{oof}}(x_j)$, the corrected prediction for a query q is

$$\hat{y}(q) = s(q) + \alpha \, \frac{\sum_{j \in \mathcal{N}_k(q)} w_{qj} \, r_j}{\sum_{j \in \mathcal{N}_k(q)} w_{qj} + \kappa}, \qquad w_{qj} = \exp\big(-(d(q,j)/h)^2\big), \tag{1}$$

where $\mathcal{N}_k(q)$ are the k nearest training compositions under a composition distance d (Euclidean on standardised Magpie vectors, or L1 on element fractions). The pseudo-count κ provides the novelty damping: when all neighbours are far, $\sum w$ is small and the correction vanishes. For classification the shift is applied on the logit scale with probability residuals. The tuple $(d, k, h, \kappa, \alpha)$ is selected on the training rows only, by scoring the corrected OOF predictions (leave-self-out neighbour sets) with the task metric over a fixed grid: $k \in \{5, 10, 20, 40\}$, h a multiple $\{0.5, 1, 2, 4, 8, 16\}$ of the median nearest-neighbour distance, $\kappa \in \{0, 0.1, 0.5, 2\}$, and $\alpha \in \{0.25, 0.5, 0.75, 1\}$ (regression) or $\{0.1, 0.25, 0.5, 1, 2, 4\}$ (classification). The uncorrected stack ($\alpha = 0$) is always a candidate and wins ties, so the procedure may decline to correct.

3.5. Fold-matched correction

The deployed stack $s = F^{\text{full}}$ was trained on the very neighbours j whose OOF residuals Eq. (1) borrows. Writing $F^{(-m)}$ for the stack whose experts excluded inner fold P_m, the naive transfer decomposes as $r_j = [r_j + F^{(-m(j))}(q) - F^{\text{full}}(q)] + [F^{\text{full}}(q) - F^{(-m(j))}(q)]$. The second bracket, the *absorption term*, measures how far the refit has already moved toward the neighbour, and the naive correction counts it a second time; for interpolating experts and near-duplicate neighbours it can be as large as r_j itself. The fold-matched variant transfers only the first bracket, $T_{qj} = r_j + F^{(-m(j))}(q) - F^{\text{full}}(q)$, in place of r_j in Eq. (1). Its selection uses deployment-matched training instances: for a training row $i \in P_m$ and a neighbour $j \in P_{m'}$ ($m' \neq m$) the transfer is computed from the pair-excluded refit $F^{(-m,-m')}$, which saw neither i nor j; neighbours in the same inner fold use r_j directly. This requires $J(J-1)/2 = 10$ extra fits per expert and fold, with the stack weights held fixed. The variant was proposed by a structured ideation pipeline (ResearchStudio IdeaSpark), whose coherence check verified the algebraic decomposition on synthetic instances; we pre-registered it as a secondary arm, not as the primary method. Relative to that proposal we simplified the damping map to the same (α, κ) family as Eq. (1), used the same two distances instead of the element mover's distance, and did not add TabPFN or MODNet experts, for reasons of CPU cost.

3.6. Hard labels for classification

Most classification entries on the Matbench leaderboard report a "ROC-AUC" equal to their balanced accuracy, which is what `matbench` computes when hard labels rather than probabilities are recorded. For a like-for-like comparison we therefore also report balanced accuracy at a threshold chosen inside each training fold (maximising balanced accuracy on the arm's training OOF predictions), alongside the probability ROC-AUC.

3.7. Protocol, pre-registration and statistics

All model and hyperparameter selection was done by cross-validation within each official training fold. Concretely, the candidate design configurations (with or without the 21 extra descriptors; L2 or L1 stacking loss; with or without an L1-objective LightGBM expert) were first explored by inner cross-validation on the training rows of fold 0 only. Then, inside every official training fold T_k, all candidates were cross-fitted and the one with the best cross-fitted OOF score of the uncorrected global stack was kept; because

this criterion uses the uncorrected stack, it favours neither correction arm. The correction hyperparameters were then selected inside T_k as described above. The analysis plan, including an amendment that introduced this per-fold selection and was written before any official test fold was scored, and the SHA-256 hashes of all code were committed before the final run; both are provided in the repository and Supplementary Material. (An earlier run under the original plan, in which the design was fixed on fold 0's training rows, was stopped and never scored.)

The primary endpoint, per task, is Δ = mean over the five official folds of the metric of the corrected stack minus that of the uncorrected stack (MAE for regression, probability ROC-AUC for classification). Its test is a paired bootstrap that resamples test points within folds (B = 10,000), with Holm correction across the four tasks; a task counts as improved only if Δ favours the correction and the Holm-adjusted $p < 0.05$. Because five folds give little power, we also report per-fold differences, the number of folds improved, and the Nadeau–Bengio corrected resampled t-test (Nadeau and Bengio, 2003). Secondary comparisons (fold-matched versus stack and versus naive, stack versus the best single expert selected by inner CV, kernel-space ablations, correction without stacking) and analyses (errors stratified by distance to the nearest training composition, calibration, selected configurations, and a label-permutation control) were also pre-specified.

4. Results

4.1. All arms on the official folds

Table 1 lists every arm on every task (mean ± standard deviation over the five official folds), and Table 2 the pre-registered tests. The equal-weight mean of the experts is clearly the worst ensemble on all tasks; the simplex stack never loses to it. Against the best single expert chosen by inner cross-validation, the stack is better on glass (+0.20% ROC-AUC, $p_{\mathrm{boot}} = 0.002$, 5/5 folds) and is within noise on the other three tasks (steels -0.35%, expt_gap +0.13%, expt_is_metal +0.06%). The extremely randomised trees expert carries most of the stack weight on the two regression tasks, and LightGBM and extra trees share it on the classification tasks (Table 6).

4.2. Primary endpoint: does the local correction transfer?

The correction improved 1 of the four tasks under the pre-registered rule. On expt_gap the corrected stack reached an MAE of 0.3240 eV against

Table 1: Every arm on the four Matbench composition tasks (official 5-fold CV; mean ± s.d. over folds). Regression: MAE (lower is better); classification: ROC-AUC from probabilities (higher is better). Single-expert rows are means over the folds in which the expert was part of the selected configuration.

Arm	steels MAE (MPa)	expt_gap MAE (eV)	expt_is_metal ROC-AUC	glass ROC-AUC
Best single (inner CV)	83.41 ± 10.86	0.3259 ± 0.0130	0.9819 ± 0.0015	0.9548 ± 0.0097
Equal-weight mean	102.46 ± 20.11	0.4007 ± 0.0242	0.9811 ± 0.0027	0.9458 ± 0.0116
Simplex stack	83.12 ± 10.89	0.3264 ± 0.0139	0.9825 ± 0.0017	0.9567 ± 0.0091
Stack + local corr. (naive)	82.80 ± 11.65	0.3240 ± 0.0140	0.9825 ± 0.0018	0.9567 ± 0.0091
ablation: Magpie space only	83.53 ± 11.30	0.3235 ± 0.0139	0.9825 ± 0.0017	0.9567 ± 0.0091
ablation: fraction space only	82.80 ± 11.65	0.3235 ± 0.0142	0.9825 ± 0.0018	0.9567 ± 0.0091
Stack + fold-matched corr.	82.02 ± 9.12	0.3276 ± 0.0149	0.9826 ± 0.0017	0.9559 ± 0.0094
Best single + local corr.	82.69 ± 11.60	0.3239 ± 0.0133	0.9819 ± 0.0015	0.9548 ± 0.0097
Expert et_all	83.41	0.3259	0.9809	0.9548
Expert krr_frac	160.38	0.5613	–	–
Expert krr_magpie	109.37	0.4840	–	–
Expert lgbm_all	88.00	0.3425	0.9819	0.9519
Expert lgbm_l1_all	93.97	–	–	–
Expert mlp_all	156.17	0.4611	0.9702	0.9055
Expert rf_all	96.62	0.3645	0.9808	0.9516
Expert svc_frac	–	–	0.9472	0.8735
Expert svc_magpie	–	–	0.9566	0.9009

0.3264 eV for the uncorrected stack ($\Delta = -0.00232$ eV, -0.71%), with every one of the five folds improved and $p_{\text{Holm}} = 0.005$ (Nadeau–Bengio $p = 0.011$). The effect is real but small: about 0.7% of the MAE. On steels the change was -0.39% ($p_{\text{Holm}} = 1.000$, 3/5 folds improved), indistinguishable from zero given a fold-to-fold standard deviation of about 10.89 MPa. On the two classification tasks the inner-CV selection almost never switched the correction on (glass: 0/5 folds; expt_is_metal: 1/5), so the corrected and uncorrected stacks are identical or nearly so (Δ ROC-AUC $= -0.00000$ and $+0.00000$). In short, the correction that was confirmed in our chemistry study transfers only weakly: it gives a consistent sub-percent gain on the largest regression task, nothing measurable on steels, and is (correctly) declined by its own selection procedure on the classification tasks.

4.3. Where the correction helps and hurts

Fig. 1 stratifies the test errors by the distance from each test composition to its nearest training composition. On both regression tasks the naive correction *hurt* in the nearest-support quartile (expt_gap +0.9%, steels +6.8% MAE relative to the stack) and helped in the other three (expt_gap -1.0%,

Table 2: Pre-registered primary test (naive local correction vs. uncorrected stack) and key secondary comparisons. Δ is the mean over folds of the metric difference (MAE for regression, where negative favours A; ROC-AUC for classification, where positive favours A). CI and p_{boot}: paired bootstrap over test points within folds ($B = 10^4$); p_{Holm}: Holm correction over the four primary tests; p_{NB}: Nadeau–Bengio corrected resampled t-test over the five folds.

Task	Comparison (A vs B)	Δ = A−B (rel.)	95% CI	p_{boot}	p_{Holm}	folds A better	p_{NB}
steels	naive local vs stack (primary)	-0.3202 (-0.39%)	[-2.15, +1.5]	0.720	1.000	3/5	0.896
steels	naive local vs best single	-0.6135 (-0.74%)	[-2.63, +1.34]	0.538	–	3/5	0.816
steels	stack vs best single	-0.2933 (-0.35%)	[-1.65, +1.08]	0.660	–	2/5	0.417
steels	fold-matched vs stack	-1.097 (-1.32%)	[-4.32, +2.13]	0.519	–	2/5	0.817
steels	fold-matched vs naive local	-0.7765 (-0.94%)	[-3.45, +1.91]	0.583	–	3/5	0.812
expt_gap	naive local vs stack (primary)	-0.00232 (-0.71%)	[-0.0037, -0.000942]	0.001	0.005	5/5	0.011
expt_gap	naive local vs best single	-0.001895 (-0.58%)	[-0.00391, +0.000155]	0.071	–	4/5	0.108
expt_gap	stack vs best single	+0.0004248 (+0.13%)	[-0.000988, +0.00181]	0.566	–	3/5	0.695
expt_gap	fold-matched vs stack	+0.001241 (+0.38%)	[-0.00145, +0.00418]	0.390	–	2/5	0.516
expt_gap	fold-matched vs naive local	+0.00356 (+1.10%)	[+0.0012, +0.00602]	0.005	–	0/5	0.096
expt_is_metal	naive local vs stack (primary)	-3.305e-06 (-0.00%)	[-4.88e-05, +4.3e-05]	0.899	1.000	0/5	0.541
expt_is_metal	naive local vs best single	+0.0005853 (+0.06%)	[-0.000504, +0.0017]	0.301	–	4/5	0.348
expt_is_metal	stack vs best single	+0.0005886 (+0.06%)	[-0.000507, +0.0017]	0.275	–	4/5	0.345
expt_is_metal	fold-matched vs stack	+0.0001571 (+0.02%)	[-0.000148, +0.00047]	0.320	–	3/5	0.577
expt_is_metal	fold-matched vs naive local	+0.0001604 (+0.02%)	[-0.000128, +0.000458]	0.277	–	3/5	0.566
glass	naive local vs stack (primary)	+0 (+0.00%)	[+0, +0]	1.000	1.000	0/5	1.000
glass	naive local vs best single	+0.001897 (+0.20%)	[+0.000704, +0.00314]	0.002	–	5/5	0.141
glass	stack vs best single	+0.001897 (+0.20%)	[+0.000689, +0.00314]	0.002	–	5/5	0.141
glass	fold-matched vs stack	-0.0008173 (-0.09%)	[-0.00194, +0.000315]	0.153	–	1/5	0.247
glass	fold-matched vs naive local	-0.0008173 (-0.09%)	[-0.00192, +0.000273]	0.149	–	1/5	0.247

-1.1%, -0.9%). That is the pattern the fold-matched analysis predicts: for a near-duplicate neighbour the deployed stack has already absorbed the neighbour's error, so adding its OOF residual counts it twice. The fold-matched variant, however, did not repair it: on expt_gap it was worse than the naive correction overall (+1.10% MAE, $p_{\text{boot}} = 0.005$, 0/5 folds better) and also worse in the nearest quartile (+2.0%). On steels it gave the lowest mean MAE of all arms (82.02 MPa) but with no detectable difference from the stack (-1.32%, $p_{\text{boot}} = 0.519$). On the classification tasks it was switched on in every fold but changed ROC-AUC by less than 0.1% in either direction.

4.4. Ablations and controls

Restricting the neighbour search to one composition space gave essentially the same gain on expt_gap (Magpie -0.86%, element fractions -0.88%), whereas on steels only the element-fraction space was selected in practice and the Magpie-only variant was slightly worse than the stack (+0.50%, 0/5 folds better). Applying the correction to the best single expert instead of the stack ("correction without stacking") improved expt_gap by -0.63% relative to that expert ($p_{\text{boot}} = 0.002$, 5/5 folds), so on this task the gain does not depend on

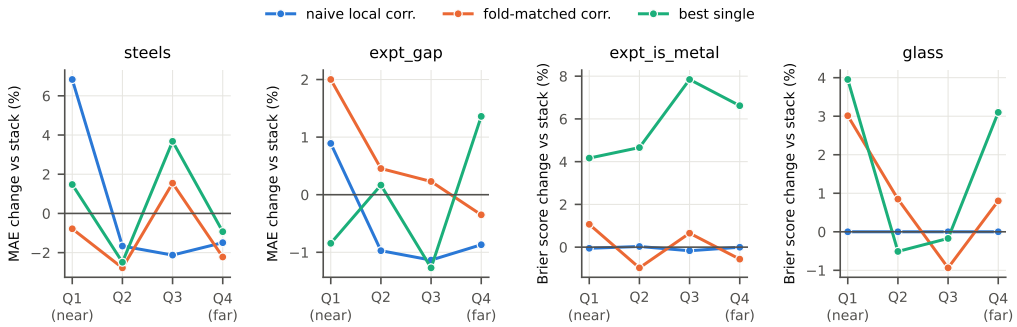


Figure 1: Error change relative to the uncorrected stack, by support quartile (distance from the test composition to its nearest training composition; bin edges are the training rows' leave-one-out quartiles in each fold). Regression: MAE; classification: Brier score. Negative is better.

Table 3: Number of official folds (of 5) in which the inner-CV selection switched each correction on, and the same for the label-permutation control.

Task	naive corr. on	fold-matched corr. on	perm. control on (naive)	perm. control on (f.-m.)
steels	5/5	5/5	5/5	5/5
expt_gap	5/5	5/5	5/5	5/5
expt_is_metal	1/5	5/5	5/5	5/5
glass	0/5	5/5	5/5	5/5

stacking. The label-permutation control did not behave as a clean null: with labels permuted within inner folds the selection still switched a correction on in every fold of every task (Table 3), because shrinking predictions toward a neighbourhood mean lowers the training loss against permuted labels too. We pre-registered this control as diagnostic only, and it shows that a non-zero selected strength is not by itself evidence of local structure.

4.5. Comparison with the Matbench leaderboard

Table 4 places the pre-registered primary arm on the leaderboard snapshot of 18 October 2026. For the classification tasks we compare balanced accuracy at the inner-CV threshold with the board's balanced-accuracy column, because the board's "ROC-AUC" values of the label-based entries equal their balanced accuracy. The method would rank 3 of 12 on steels (behind TPOT-Mat and AutoML-Mat, ahead of MODNet v0.1.12), 2 of 13 on expt_gap (behind Darwin, ahead of the CrabNet and MODNet entries), 2 of 8 on expt_is_metal and 2 of 8 on glass. It does not beat the best entry on

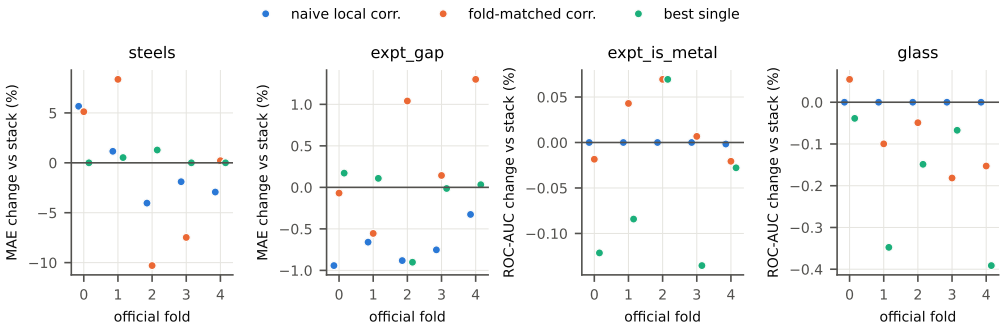


Figure 2: Per-fold change of the test metric relative to the uncorrected stack for the naive and fold-matched corrections and the best single expert (selected by inner CV). For classification a positive change in ROC-AUC is better.

Table 4: Primary arm versus the Matbench v0.1 leaderboard (snapshot 18 October 2026). Classification: balanced accuracy at the threshold chosen inside each training fold, compared with the board's balanced-accuracy column. Rank counts our entry among the board entries.

Task	Metric	Ours: stack + naive corr.	Best board entry	Rank
steels	MAE	82.80 ± 11.65	TPOT-Mat: 79.95 ± 13.59	3 of 12
expt_gap	MAE	0.3240 ± 0.0140	Darwin: 0.2865 ± 0.0083	2 of 13
expt_is_metal	balanced acc.	0.9370 ± 0.0063	Darwin: 0.9598	2 of 8
glass	balanced acc.	0.8841 ± 0.0184	MODNet (v0.1.12): 0.9603	2 of 8

any task. On expt_gap and glass the gaps to the leaders are large (0.3240 vs. 0.2865 eV; balanced accuracy 0.8841 vs. 0.9603), and on expt_is_metal it is 0.023 in balanced accuracy. Our probability ROC-AUCs (0.9825 on expt_is_metal, 0.9567 on glass) are not comparable with the label-based board values and are not ranked.

4.6. Calibration and compute

Stacking improved the Brier score and the expected calibration error over the best single expert on both classification tasks, and the corrections left them essentially unchanged (Table 5). On 24 CPU cores (5 parallel workers of 4 threads) one official fold took a median of about 1, 16, 8 and 44 minutes for steels, expt_gap, expt_is_metal and glass when every expert was computed from scratch (the glass figure includes a system-standby pause); the pair-excluded refits of the fold-matched arm account for 44%, 33% and 37% of the expert-fitting time on the first three tasks (84% on glass, inflated by the

Table 5: Calibration on the classification tasks (means over folds) and balanced accuracy at the inner-CV threshold.

Task	Arm	Brier	log loss	ECE	bal. acc.
expt_is_metal	Best single	0.0498	0.2007	0.0369	0.9378
expt_is_metal	Stack	0.0469	0.1662	0.0201	0.9374
expt_is_metal	Stack + naive corr.	0.0468	0.1662	0.0192	0.9370
expt_is_metal	Stack + fold-matched corr.	0.0469	0.1660	0.0211	0.9366
glass	Best single	0.0747	0.2451	0.0220	0.8827
glass	Stack	0.0735	0.2398	0.0211	0.8841
glass	Stack + naive corr.	0.0735	0.2398	0.0211	0.8841
glass	Stack + fold-matched corr.	0.0740	0.2411	0.0232	0.8835

Table 6: Mean simplex-stack weight per expert (over folds; "–" where the expert was not in the selected configuration of every fold).

Expert	steels	expt_gap	expt_is_metal	glass
et_all	0.94	0.86	0.41	0.57
krr_frac	0.00	0.00	–	–
krr_magpie	0.06	0.00	–	–
lgbm_all	0.00	0.14	0.48	0.27
lgbm_l1_all	–	–	–	–
mlp_all	0.00	0.00	0.05	0.00
rf_all	0.00	0.00	0.03	0.10
svc_frac	–	–	0.01	0.06
svc_magpie	–	–	0.02	0.00

same pause).

5. Discussion

What transferred. The cross-fitted simplex stack is a sound default on these tasks. It never lost to an equal-weight ensemble, it improved calibration on both classification tasks, and it was the best probabilistic classifier on glass. On the other tasks it was statistically indistinguishable from the best single expert, which is itself selected by inner CV. The novelty-damped local OOF correction, the part that SIG-Net confirmed for chemical properties, transferred only partly. It produced a consistent but small gain on the experimental band gap: all five folds improved, by 0.7% of the MAE, and the gain

survives Holm correction. It made no measurable difference on steels. On the two classification tasks its own inner-CV selection almost never switched it on. We read the band-gap result as a genuine but modest effect. It is of the size that the honesty rules of this study led us to expect, and much smaller than what is needed to close the gaps to the leaderboard leaders.

Why it does not do more. The support-stratified analysis points to a mechanism. On both regression tasks the naive correction is harmful for test compositions that have a near-duplicate in the training set. This is where the deployed stack has already absorbed the neighbour's error, and adding the neighbour's OOF residual double-counts it. The correction helps only at intermediate and large distances. The fold-matched variant was designed to remove exactly this absorption term, but it did not help in practice. One possible reason: the pair-excluded refits are trained on 60% of the fold, and on these data that appears to make the transferred analogue estimates noisier than the bias they remove. We regard this as a negative result for that proposal, not a refutation of the decomposition, which is an algebraic identity. The label-permutation control also shows that selecting a non-zero correction strength is weak evidence of local structure, because shrinkage toward a local mean pays off even against permuted labels. A stricter selection rule, for example a one-standard-error rule over inner folds, might switch the correction off more often. It would not have changed the qualitative conclusions here.

The leaderboard. With CPU-only, off-the-shelf experts and fully nested selection, the method would rank 3, 2, 2 and 2 on steels, expt_gap, expt_is_metal and glass among the leaderboard entries of 18 October 2026. It is never first. The gaps to Darwin on expt_gap and to MODNet v0.1.12 on glass are far larger than anything a residual correction contributed here. Closing them would need better base representations, not a better correction stage. For the classification tasks we found that most board entries recorded hard labels, so their "ROC-AUC" is a balanced accuracy. Comparing probability ROC-AUCs with these values would have inflated our standing considerably, for example 0.9825 versus 0.9598 on expt_is_metal. We recommend that future Matbench classification submissions report both quantities explicitly.

Limitations. The study covers four composition tasks only. The expert pool is restricted to CPU-feasible models (no CrabNet, Roost, MODNet or TabPFN experts), and five official folds give little power for per-task comparisons; the Nadeau–Bengio p-values in Table 2 show how weak fold-level

evidence is. The set of candidate design configurations was itself constructed while looking at inner-CV results on fold 0's training rows; only the choice among the candidates was made inside every training fold. We did not tune the expert hyperparameters per fold beyond the internal kernel-ridge grid, and we used a single seed. The fold-matched arm was implemented in a simplified form of the proposal: the same damping family as the naive correction, two composition distances, and no element mover's distance. The ideation, prior-art and review steps that suggested it were partly automated, and their full records are in the repository.

6. Conclusions

Under a pre-registered, fully nested protocol on the official Matbench folds, cross-fitted stacking of diverse composition experts is a reliable baseline. The novelty-damped local OOF residual correction transfers only weakly from chemistry to materials: a 0.7% MAE gain on the experimental band gap, and none elsewhere. The correction hurts for near-duplicate compositions. A fold-matched variant aimed at that failure did not fix it. We report these mixed results in full, together with every arm, fold and selection, as a calibrated data point for anyone tempted to bolt a residual-correction stage onto a materials model.

Data and code availability

All code, the frozen analysis plan (`ANALYSIS_PLAN.md`), SHA-256 freeze hashes, per-fold predictions of every arm, per-fold metrics and the unsubmitted Matbench result files are in the repository `https://github.com/bhaskargurram-ai/matbench-composition-stack`, which will be made public on submission. The data are the public Matbench v0.1 tasks.

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CRediT authorship contribution statement

Bhaskar Gurram: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

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The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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During the preparation of this work the author used Claude (Anthropic), running the ResearchStudio idea-generation skills, to assist with literature search, prior-art checking, code development and drafting of the manuscript. All analysis code, results and text were reviewed by the author, who takes full responsibility for the content of the publication.

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